Documentation

This project was built with a surface EMG measurement, using bipolar sensors. The sensors were run on the STM32 Nucleo F446ZE. We used the Ylab studio environment and the Ylab zero program to process the EMG. For a detailed explanation on the sensor placement and data processing pipeline, please refer to the methods. Due to inaccessibility of the LapSim simulator, a simple box-simulator substitute could be used. Please see the replication section for a detailed report. Below, we present several photos to visually introduce the idea and setup using the LapSim simulator (see Figures 1 through 7). For a video demonstration, please see the Github repository by SVos03.

Figure 1

Performance rate for instrument navigation task

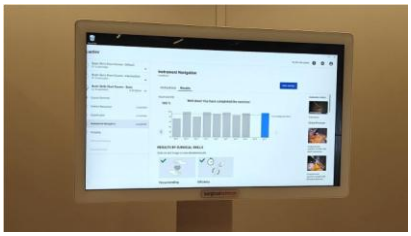


Figure 2

Demonstration of the participant performing the task

Figure 3

Demonstration of sensor placement for flexor and extensor digitorum

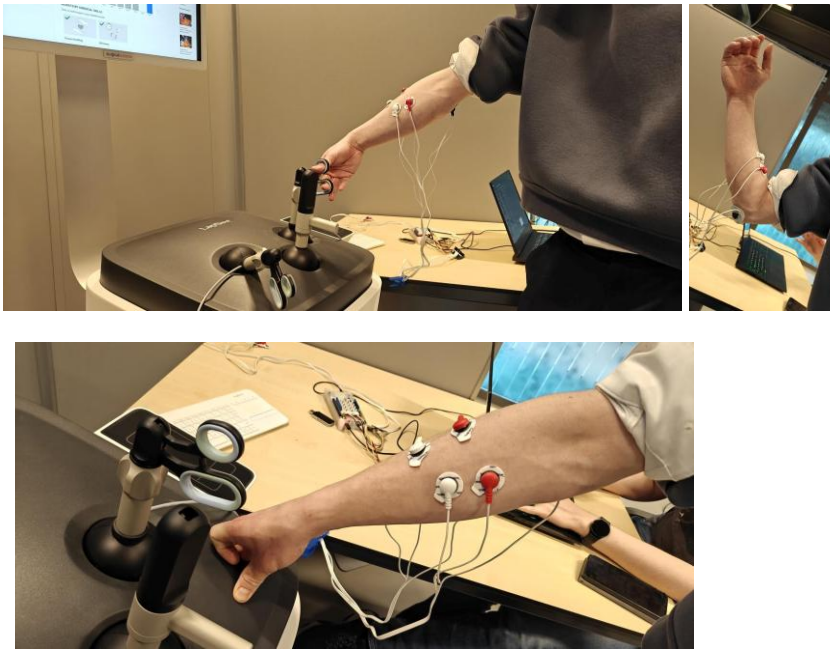
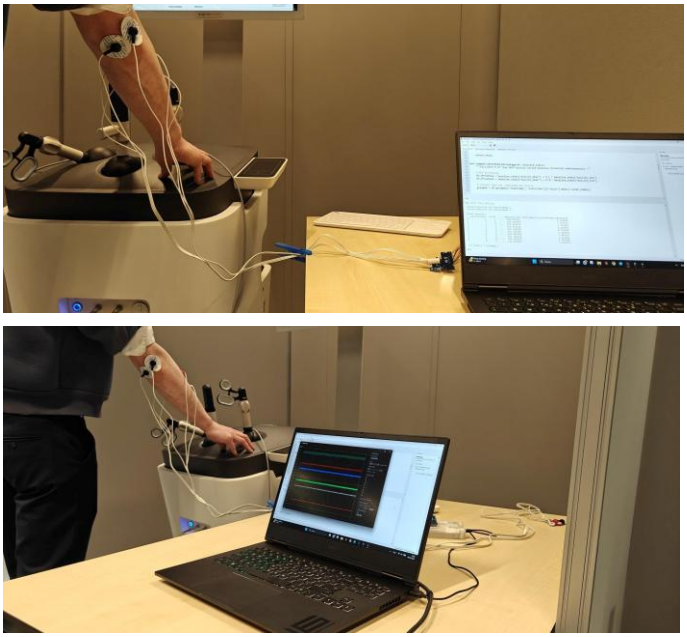


Figure 4

Demonstration of data processing



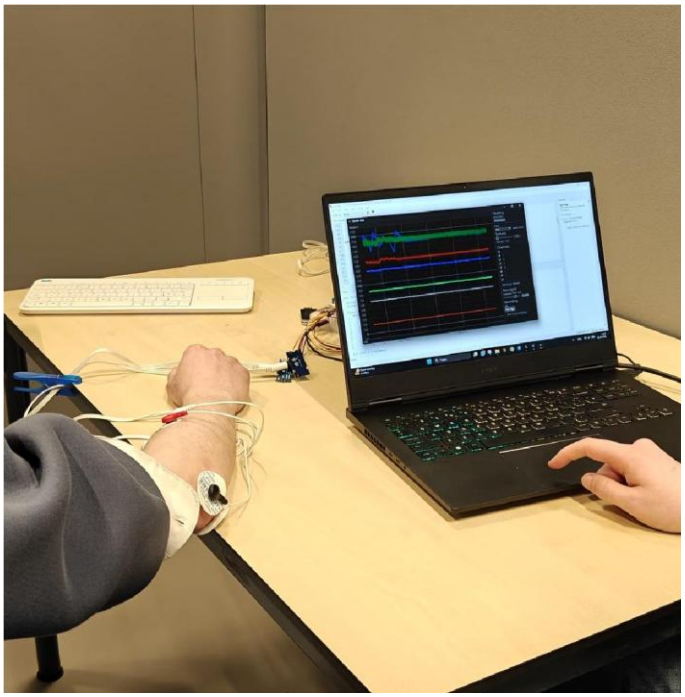


Figure 5

Human Factors considerations (cable clipper and high table)

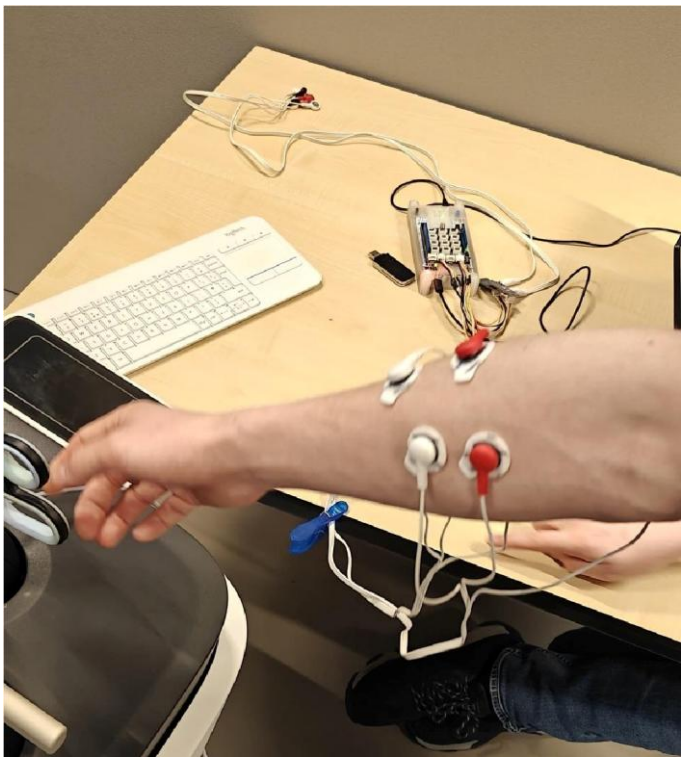
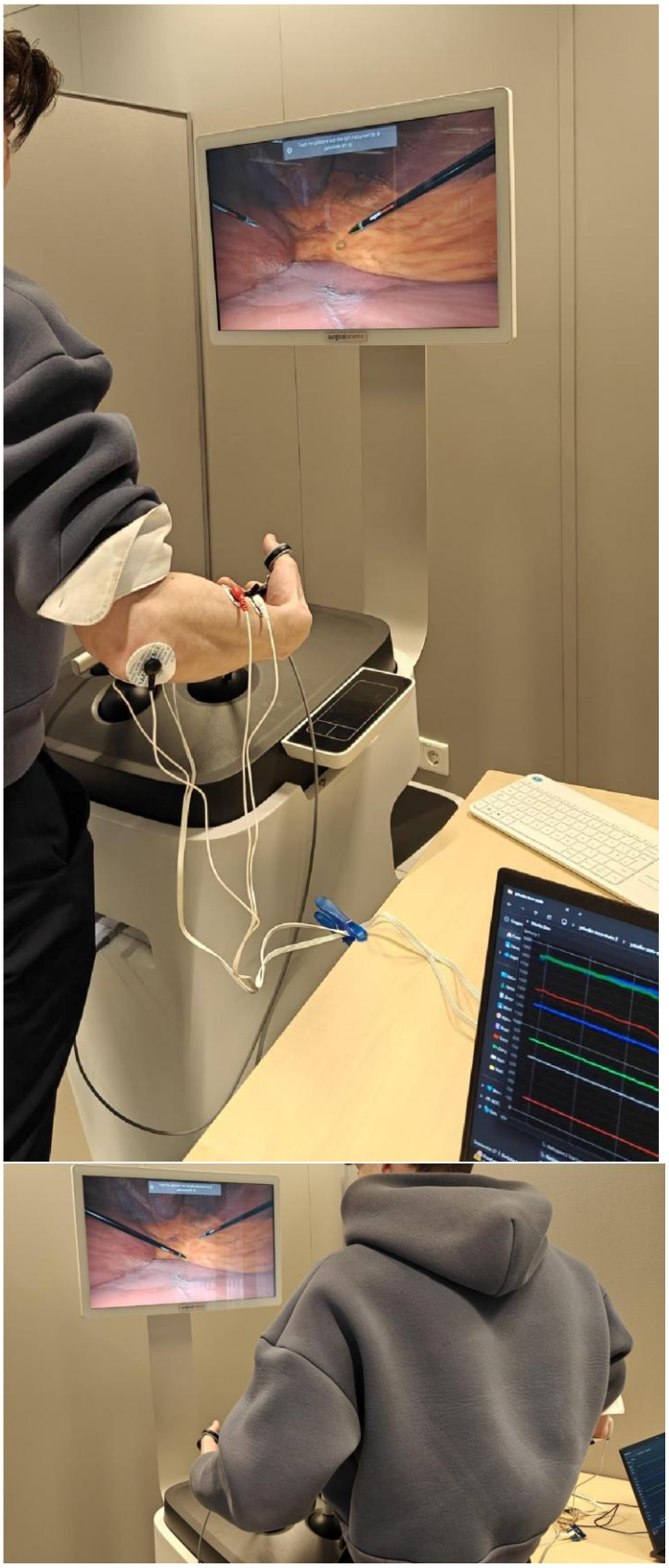


Figure 6

Demonstration of task procedure



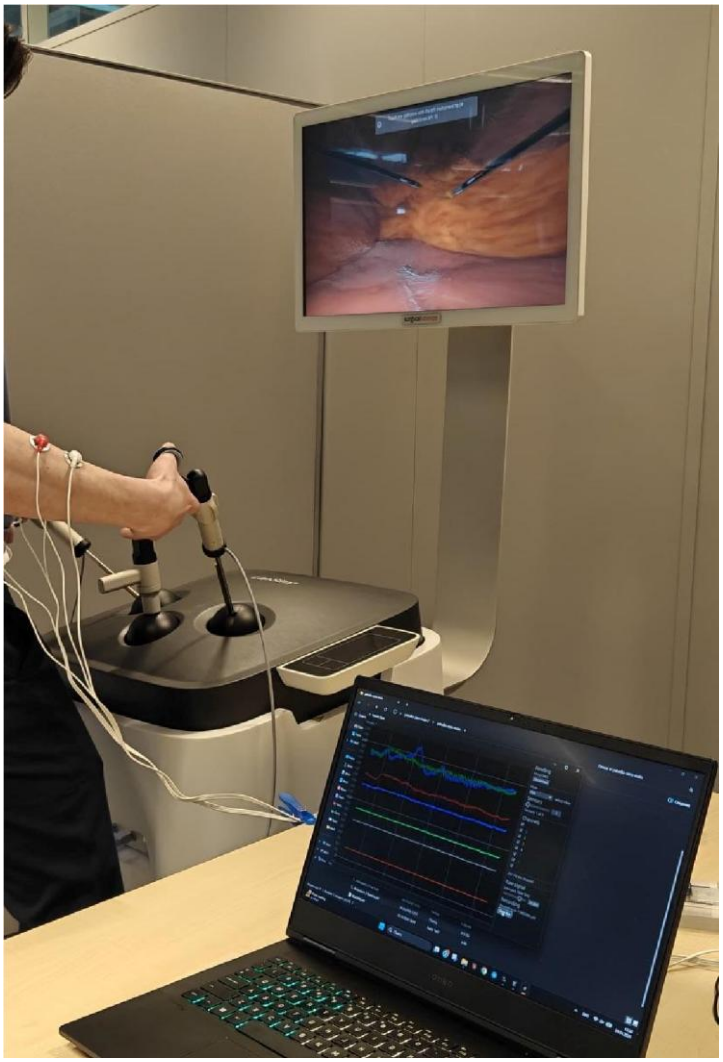
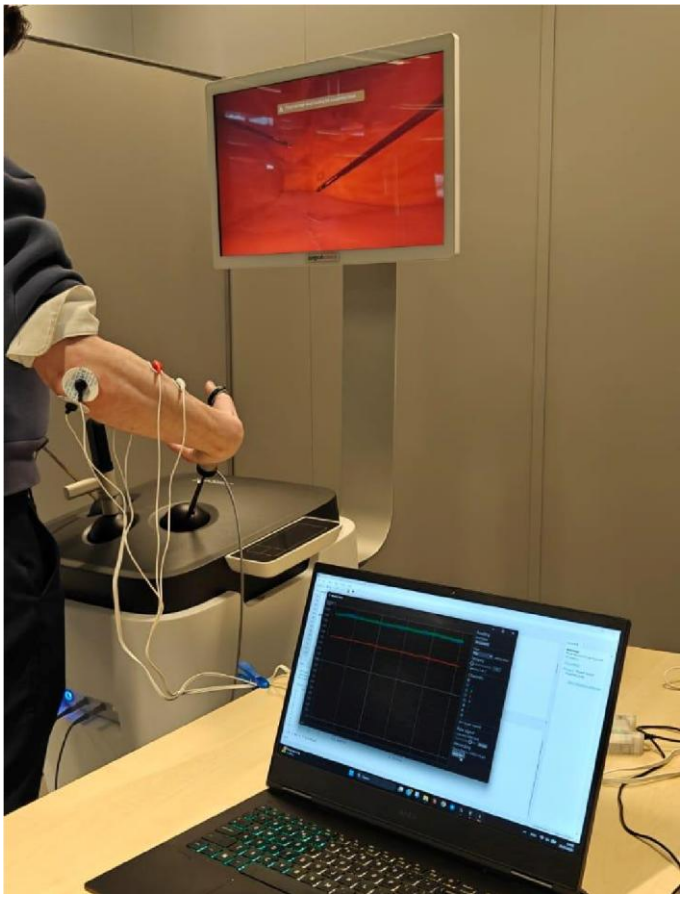
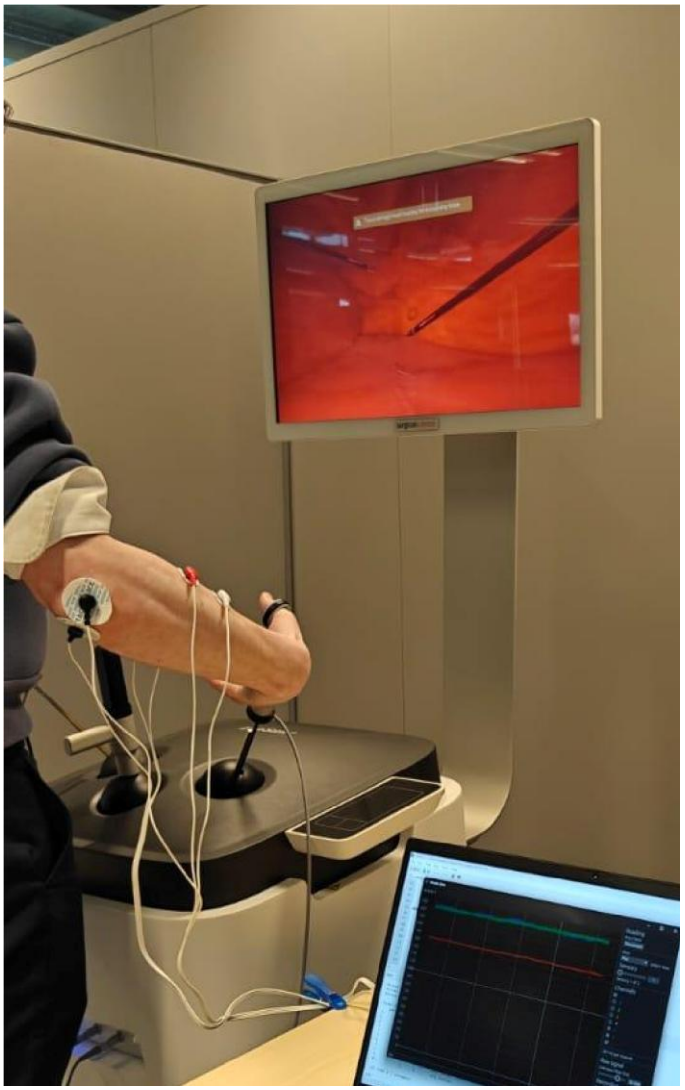


Figure 7

Demonstration of error during task performance